# Risk Assessment and Prediction of Floods and Droughts using Attention-LSTM

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***Abstract*—Floods and Droughts are among the most destructive natural disasters which cause extensive damage to infrastructure, displace populations and are threatening to life and property in a serious way. These disasters need timely and accurate prediction for the purpose of disaster management and mitigation efforts. This paper presents a new framework for the assessment and prediction of floods and droughts. The data integrated for this methodology include satellite imagery of Sentinel-1, Sentinel-2 and MODIS, weather data and historical disaster records. This paper makes a significant contribution to the field, as it presents a holistic solution for the assessment and prediction of floods and droughts. It integrates Attention Long Short Term Memory (LSTM) based model for prediction of disasters. It is the application of advanced technologies to build more resilient communities and enhance disaster preparedness. Focusing on floods and droughts, the system tries to solve two of the biggest problems in disaster management, and assists emergency responders, policymakers and communities make informed decisions and reduce the devastating consequences of these natural disasters.**

***Keywords*—Disaster Management, Floods, Droughts, Disaster assessment, Disaster prediction, Geographic Information System, Long Short-Term Memory, Raster data, Vector data**

## I. Introduction

India faces significant challenges from natural disasters such as floods and droughts due to the country's diverse climate and geographical territories. The heavy rainfall in states like Assam, Bihar and Kerala, often cause devastating floods leading to loss of lives and displacement of people. On the other hand, erratic nature of monsoon lead to severe droughts in areas such as Maharashtra, Rajasthan and Karnataka. It adversely affects the agriculture and water availability. These disasters affect people and assets due to lack of disaster management. These problems require a data-driven and technology based approach that is useful in the early identification of such risks.

This research paper presents risk assessment and prediction of floods and droughts using Attention-LSTM, a new initiative towards making improved disaster management in India. It uses advanced geospatial data, machine learning, and real-time analytics to provide accurate assessment and forecasting of disasters. Using Sentinel-1, Sentinel-2 and MODIS satellite imagery processed through Google Earth Engine, the system provides high-resolution maps of disaster affected regions. It uses the Open Meteo API to fetch historical and real-time weather information. The project uses an attention LSTM model to predict floods and droughts based on this data. It changes the face of disaster management, enabling policymakers in decision making, disaster risk mitigations and optimal resources management.

The paper further discusses an overview of existing system. Thereafter the system architecture will be presented. Under results and analysis the performance evaluation shall be given a critical dimension. Finally, the conclusion and future scope will be discussed and highlighted to ensure the impact and potential for the system to create improvements in future disaster management practices.

## II. Related Works

In this section, we discuss relevant existing research works on flood and drought assessment and prediction, and Attention LSTM model based on insight from the works cited to position our work within the relevant context.

Flood prediction and assessment have been improved using different types of satellite data and machine learning algorithms. Asiya Begum et al.(2022) [1] used satellite data for flood mapping and monitoring in Uttar Pradesh. They also provided a structured approach towards disaster assessment. According to Nugroho et al. (2021), the dataset collected from Sentinel-1 and Sentinel-2 well defined the affected area with improved accuracy [2]. Syeed et al.

(2022) had promising results using ML models in flood forecasting but faced difficulties concerning computational efficiency [3]. Similarly, Puttinaovarat and Horkaew (2020) also made use of the crowdsourced dataset, and due to the deletion of inaccurate input, there were better predictive abilities of data inconsistencies. However, the issues of false positives and data imbalance remain major challenges [4].

Orimoloye et al.(2021) applied MODIS indicator to monitor water scarcity and provided the needed information for environmental conservation [5]. Sazib et al. (2018) applied Google Earth Engine in combination with global soil moisture to accurately assess drought conditions [6]. Analysis and discussion on diverse machine learning models for drought forecasting and improvement in the predictability of drought can be seen from Mosavi and Ardabili (2023) [7]. Likewise, Zhao et al. (2023) utilised different techniques of deep learning for accuracy improvements in drought forecast [8].

Advanced machine learning models have improved weather prediction. Nitesh et al. (2023) have used long short-term memory models for forecasting which can effectively process time series data but they lack the reliability of accuracy in longer terms [9]. Further Wen and Li (2023) extended time series prediction by establishing the attention mechanism on an LSTM-Attention LSTM model that yields more accurate information about the patterns and time dependency [10].

## III. Proposed Architecture

The system addresses the growing issue of floods and droughts. It focuses on the assessment and prediction of these disasters. The system collects and uses high resolution satellite imagery for assessment and historical and current meteorological data for prediction. The following is a more detailed explanation of system architecture.

### *A. Assessment*

This section focuses on assessment of historical flood and drought events. It collects satellite data of Sentinel-1A, Sentinel-2 and MODIS (Moderate Resolution Imaging Spectroradiometer) from Google Earth Engine (GEE) platform. For flood assessment, radar data is used to map the flooded regions and for drought assessment the vegetation indices are used.

**I. Flood Assessment:** There are two parts in Flood assessment: flood assessment map and flood risk map. This becomes useful in disaster management, response or in carrying out the post disaster rescue. The regions which gets exposed to flooding are identified using flood assessment map. The entire process involved in Fig. 1 is covered with the help of following steps:

**a. Extraction of a shapefile:** This step deals with the extraction of a shapefile. A shapefile defines the geographic area of interest for the flood assessment.

**b. Importing Sentinel-1 Radar Imagery:** This imports radar data from Sentinal-1 satellite. This satellite uses Synthetic Aperture Radar (SAR) technology, which allows them to see through clouds and darkness.

**c. Filtering Radar Imagery for VV Polarization:** The Sentinel-1 dataset has dual polarization radar bands (VV and VH). For this analysis, VV polarization is used because it can capture water surface efficiently. This makes it useful to identify features and changes in flooded regions.

**d. Selecting and Clipping Radar Images:** The flood effect can be highlighted using before and after image of the flooded region. These images are clipped to the Area of Interest (AOI). This makes sure that the analysis is done in the affected areas which reduces computational complexity making the whole process lighter.

**e. Applying Smoothing Filter:** To reduce the noise in the radar data a smoothing filter is applied to both the pre-flood and post-flood radar images. The smoothing filter used in this case is a 30 meter median focal filter which easily distinguish between flooded areas.

**f. Calculating the Difference Between Pre and Post-Flood Radar Images:** To find the difference in backscatter values the pre-flood and post-flood images are substracted which gives us the flood affected regions. The flood affected regions are identified due their radar backscatter value being reduced by 3 dB or more.

**g. Identifying Flood Extent based on Threshold:** From difference image, the threshold -3 dB is used to identify the flood affected areas. This ensures that the permanent water bodies which have backscatter value of -20 dB or below are not mistaken as flooded regions.

**h. Displaying Map for Visualization:** In Flood Extent Map the area marked in red shows the flooded region and blue represents permanent water bodies.

The assessment using flood risk map makes use of several raster data sources which includes Sentinel 1, Sentinel 2 radar imagery, elevation data and water occurrence data from the Global Surface Water (GSW) dataset. This involves the following steps:

**a. Extraction of Vector Data:** This steps involves downloading a shapefile for the region on which we will be conducting flood assessment.

**b. Importing Raster Data:** The following raster datasets were imported: Radar Imagery from Sentinel 1, Sentinel 2, Global Surface Water (GSW) Data, SRTM Elevation Data and Landsat 8 Imagery.

**c. Water and Distance Analysis:** The water occurrence data was clipped to the area of interest. Pixels with an occurrence value above 30% were marked as permanent water bodies. A distance transform algorithm was then used to measure how far each point was from the water then these results were

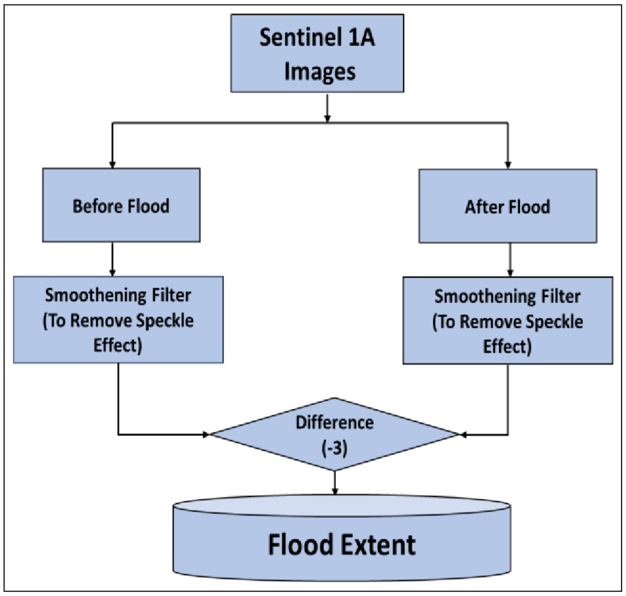


Fig. 1. Flood Assessment using Sentinel-1A satellite imagery

divided into different hazard scores in analyzing flood risk.

**d. Elevation and Topographic Analysis:** The elevation within the area of interest is retrieved using SRTM data. Lower elevated regions are marked as high-risk flood zones and vice versa. To calculate the Topographic Position Index (TPI), the elevation is subtracted from the focal mean of the Digital Elevation Model (DEM). This step helps improve the flood hazard scores for better accuracy.

**e. Vegetation and Wetness Analysis:** To find areas at risk of flooding, the NDVI (Normalized Difference Vegetation Index) is calculated using the Red and Near-Infrared bands from Sentinel-2. This helps determine vegetation cover which can act as a natural barrier against the flood. The NDWI (Normalized Difference Water Index) is also calculated to detect water presence in the landscape.

**f. Flood Hazard Scoring:** The individual hazard layers of distance, elevation, topography, vegetation, and wetness were combined to create the final flood hazard score. This score was then categorized from very low to very high using threshold values. This classification helps in a detailed flood risk assessment.

**g. Map Visualization:** The flood hazard map was displayed using a color-coded legend to identify risk levels. Areas with very high hazard were marked in red, high in yellow, medium in green, low in cyan, and very low in blue.

**II. Drought Assessment:** Google Earth Engine uses MODIS satellite imagery to assess and monitor drought conditions. MODIS data helps to map and visualize the droughts using Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI), and Land Surface Temperature (LST). The process presented in Fig. 2 is described as follows:

**a. Area of Interest (AOI):** In this study, the area is defined as feature collection and it is filtered by region name. The map was then centered on AOI so it is easier to focus the analysis on it.

**b. NDVI Data Preprocessing:** The scaled NDVI values from the MODIS NDVI data (MOD13Q1) were used. The actual scale of the NDVI values was divided by 10,000 to normalize.

**c. Temporal Filtering:** For the specific month the NDVI data was filtered for each years to calculate:

- Maximum NDVI ($\text{NDVI}_{\text{max}}$):The highest NDVI value for the month across all the years.
- Minimum NDVI ($\text{NDVI}_{\text{min}}$):The lowest NDVI value for the month across all the years.

NDVI for the specified month and year was then separately filtered and stored as $\text{NDVI}_T$.

**d. VCI Calculation:** The Vegetation Condition Index (VCI) was calculated using the formula adapted from [11]:

$$\text{VCI} = \frac{\text{NDVI}_T - \text{NDVI}_{\text{min}}}{\text{NDVI}_{\text{max}} - \text{NDVI}_{\text{min}}}$$

where:

- $\text{NDVI}_T$: The current NDVI for the specified month and year.
- $\text{NDVI}_{\text{max}}$: The maximum NDVI value for the same month across all years.
- $\text{NDVI}_{\text{min}}$: The minimum NDVI value for the same month across all years.

**e. Visualization and Categorization:** The map was enhanced with a color coded legend. It represents the five categories of the VCI: Extreme Dry, Severe Dry, Moderate Dry, Normal and Wet as it allows the visualization of the intensity of drought.

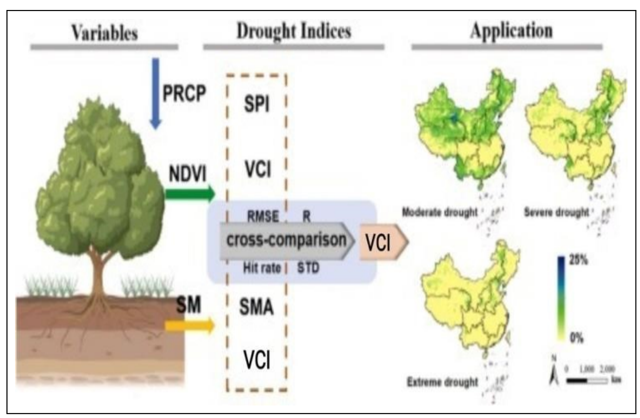


Fig. 2. Drougth Assessment Workflow

### *B. Prediction*

The deep learning models are a powerful tool for predicting floods and droughts. Accuracy can be achieved by using both historical and real-time weather and hydrological data. So the model is built using Attention LSTM network, a type of recurrent neural network designed for sequential data analysis. This allows it to recognize complex patterns and trends in climate conditions. The model is based on integrating both live weather updates and historical datasets from Open Meteo APIs which helps to make prediction resulting disaster preparedness and risk mitigation.

**a. Data Collection:** Historical datasets collected from Open Meteo API [12] plays a crucial role in identifying flood and drought conditions. For flood, the dataset collected is classified into flood and no flood data. To ensure consistency, the datasets are normalized for uniform scaling across all features. Real-time data fetched from APIs undergoes the same preprocessing to make sure it works with the trained model.

**b. Feature Selection and Data Preprocessing Techniques:**

**1. Feature selection:** From the weather data, it includes maximum temperature, minimum temperature, mean temperature, daylight duration, sunshine duration, total precipitation, total rainfall, total snowfall, precipitation hours, and maximum wind speed at 10 meters. These parameters help in analyzing weather patterns and their potential impact on floods. Additionally, from the flood data, it considers river discharge, maximum river discharge, and minimum river discharge, which are critical for assessing water flow and flood risk. From other data like date, latitude, logitude and location it extracts the month from each date and assigns a season based on predefined month ranges- Winter (December to February), Summer (March to May), Monsoon (June to September), and Post-Monsoon (October and November). This helps in analyzing seasonal trends in the data. Then it removes the columns like 'location' and 'date' to keep the dataset focused on relevant information.

**2. Data Preprocessing:**

**i. Encoding Categorical Features:** Categorical features were converted to numerical values using Label Encoding in order to make them compatible with the deep learning algorithm.

**ii. Feature Scaling:** Numeric features were standardized by MinMax Scaling within the range 0 to 1.

**iii. Handling Missing Data:** Categorical Features missing values were substituted with the most occurring value (mode) for uniformity. Numerical Features missing values were filled with the median value of the corresponding feature to minimize the effect of outliers.

**iv. Addressing Class Imbalance** The data showed class imbalance and might have had a negative impact on model performance. To balance the data Synthetic Minority Over-sampling Technique (SMOTE) was used to create artificial examples of the minority class.

**c. LSTM Model Architecture:** The LSTM network captures long dependencies in the data therefore, it is very fit for identifying the trends leading to flood or droughts (e.g., heavy rainfall and high river discharge which lead to floods, while prolonged low precipitation, less soil moisture, humidity and high temperatures lead to drought). Fig. 3 shows the working of model to predict the occurrence of flood and drought.

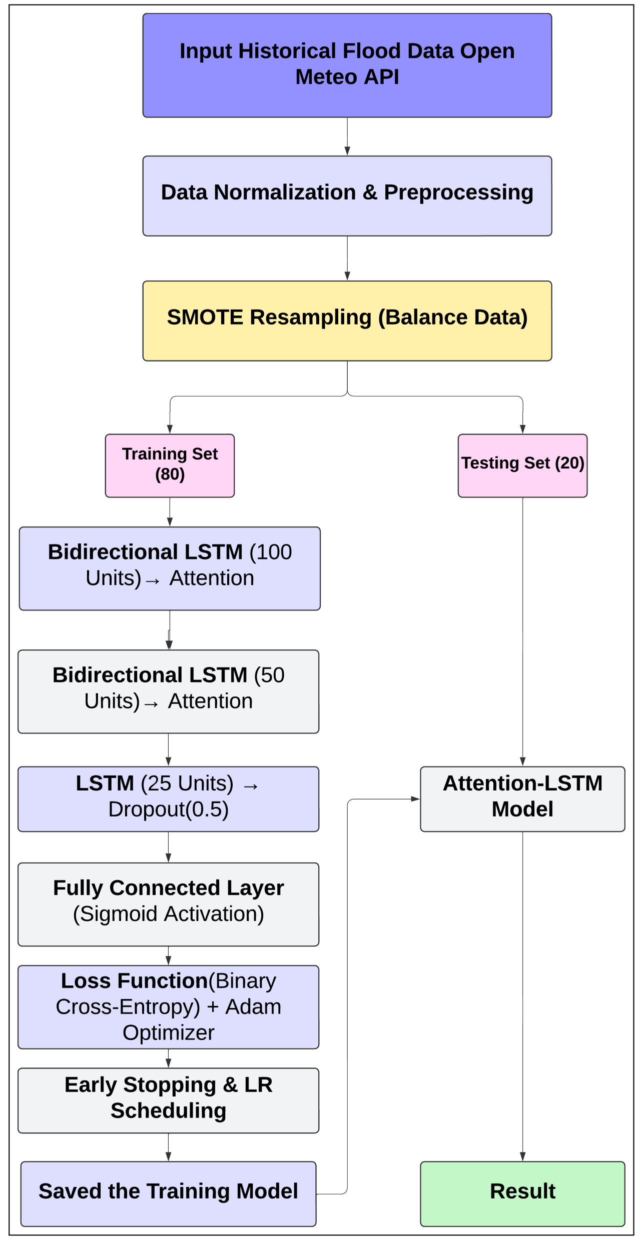


Fig. 3. Attention-LSTM Model Workflow

**d. Prediction and Alert Generation:** For flood prediction, the model computes various parameters such as precipitation, temperature and wind speed. These parameters are used to get the probability and severity of flooding. For drought prediction, it focuses on the low precipitation, low soil moisture, humidity, high temperatures, with decreased river discharge. These predictions are then compared to predetermined thresholds for risk classification and results are visualized and real-time alerts are shown.

**e. Integration and Deployment:** The model is deployed in a real-time system, where it continuously fetches live weather and hydrological data. This allows for dynamic updates of its predictions and on-time warnings for better decision-making.

## IV. Result and Analysis

The implementation of this system will significantly improve disaster awareness among general public. The system will help in early detection and assessment, making communities more prepared. The results helps in reducing emergency response time and manage floods and droughts more effectively.

### A. Dataset

The dataset used for flood assessment includes several satellite and vector datasets. The Global Surface Water Dataset, SRTM Digital Elevation Model, Sentinel-2 Surface Reflectance and Copernicus Sentinel-1 GRD with SAR imagery is used for detection of flood areas and surface. For prediction Open Meteo is used which provides weather data for flood and drought risk assessment. The data file contains in 7 days of observations from 16 locations across India. The dataset has 20 years record of flood and drought in India that provide information about climatic and hydrological interactions.

This section presents and analyzes the results of flood and drought assessment maps. Special focus is given to river discharge patterns and flood-affected areas, as they play a key role in understanding disaster impact and risks.

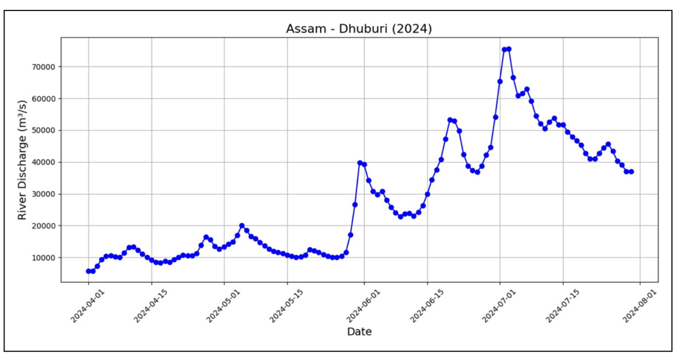


Fig. 4. Assam – Dhuburi (2024) River Discharge

The river discharge trends for Dhuburi, Assam (2024) shown in Fig. 4 are derived from Open Meteo API. Until mid-May, the rates stayed below 20,000 m³/s and reach 40,000 m³/s at the end of May. In early July it peaked at over 70,000 m³/s due to heavy rains and upstream contributions. This data is necessary for flood warnings, better planning of evacuations, and distribution of essential resources to minimize losses. Analyzing yearly trends will help in identifying the periods of high rainfall and frequent floods.

Fig. 5 identifies these risk zones in Dhuburi for July 2024 using Google Earth Engine. The affected areas were

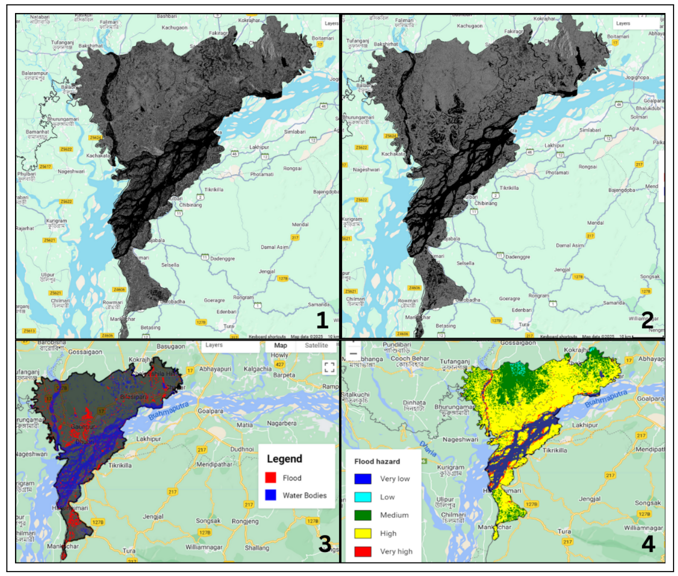


Fig. 5. Flood Assessment process for Dhubri, Assam (2024) (1) Pre-flood image (2) Post flood image (3) Post flood assessment map (4) Risk assessment map

identified by analyzing Sentinel-1 images through pre-flood [1] and post-flood [2] image differencing. The color image [3] of post flood [2] image helps to identify the flooded areas and permanent water bodies. The last one [4] uses the Global Surface Water (GSW) dataset to identify permanent water bodies and a distance transform to assess their proximity. Elevation data from SRTM DEM and TPI reveal where floods are most likely to occur. Vegetation and wetness were analyzed from Sentinel-2 satellite images where each factor (distance, elevation, topography, vegetation, and wetness) was scored from 1 to 5 to create a flood hazard index with color coded from very low to very high risk. Doing so helps the authorities to focus on areas that are at highest risk. This informs citizens about the safety of the area they reside in, hence they can better plan and store required materials and evacuate if required.

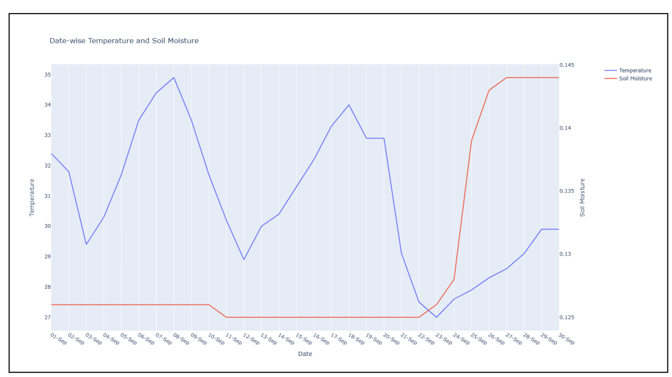


Fig. 6. Temperature vs. Soil Moisture Graph of Jaisalmer of Sept 2021

The assessment includes a graphical representation of the temperature versus soil moisture for a specific month. The graph in Fig. 6 shows the relationship between temperature and soil moisture levels during that period. It shows that increasing temperatures shows corresponding decrease in soil moisture. This month-specific analysis is useful to understand seasonal variation and it's impact on soil moisture. The drought

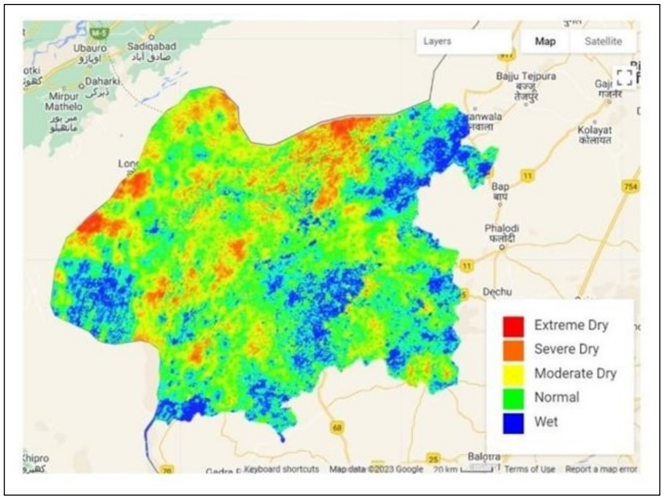


Fig. 7. Drought assessment of Jaisalmer District

assessment has also been made on specific states and their districts. The visualization approach is demonstrated in Fig. 7, which is an assessment map with different drought categories, such as extreme dry, severe dry, moderate dry, normal, and wet areas. This approach helps to identify the severity of drought conditions in the selected areas. This is crucial for efficient use of resources to reduce the effect of drought.

The performance of the Attention LSTM model was evaluated using multiple metrics, including accuracy, precision, recall and F1-score. Additionally, we compared the model with traditional forecasting models such as traditional LSTM and GRU to assess its effectiveness. The results are summarized in Table I.

TABLE I
PERFORMANCE COMPARISON OF ATTENTION-LSTM WITH TRADITIONAL MODELS

| Model | Accuracy (%) | Precision | Recall | F1-Score |
|---|---|---|---|---|
| **Attention LSTM** | 77.91 | 77.78 | 71.79 | 74.67 |
| **LSTM** | 73.26 | 66.67 | 82.05 | 73.56 |
| **GRU** | 73.26 | 69.05 | 74.36 | 71.60 |

As observed, the Attention-LSTM model outperforms LSTM and GRU in terms of accuracy, precision, recall, and F1-Score, making it a more reliable forecasting approach.

## V. CONCLUSION

The project enhances disaster management in India by using Sentinel 1, Sentinel 2 and MODIS satellite data for assessment. It uses Attention LSTM based machine learning model for prediction of floods and droughts. Given the increase in occurrence of different types of natural disasters, it is important to expand the purpose of this system to cover more types of disasters such as hurricanes, cyclones and earthquakes. Expanding its focus beyond floods and droughts, the system tries to develop new predictive models that can predict these additional disasters with higher accuracy and speed. Additionally, it also acknowledges and prioritizes the significance of the disaster recovery processes that occur after the disaster. As part of its future scope, it also seeks to incorporate robust disaster recovery measures into its system to enable quick and integrated response measures. This helps reduce the societal impacts and help in quick rehabilitation of affected areas. This paper contributes to the development of the system as a comprehensive and effective instrument for disaster management, building on innovation to help communities and societies withstand the impact of natural disasters.

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